

Rohit Poudel

Full-Stack Developer & AI / ML Engineer

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Full-stack developer and AI/ML builder based in Kathmandu, Nepal, focused on civic-tech and socially useful applications. I combine large language models, machine learning, and clean web interfaces to solve real problems — from routing citizen complaints to predicting earthquake damage. I lead VagaBond, a three-person hackathon and project team, and care more about shipping complete, polished products than one-off demos.

AREAS OF EXPERTISE

AI / ML & Data

- Python, scikit-learn, XGBoost
- LLM integration (Llama 3.3, Gemini)
- Sentence-transformers, embeddings
- UMAP / HDBSCAN clustering

Full-Stack Web & App

- Next.js, React, MERN stack
- FastAPI, Flask, Node
- Tailwind, Framer Motion
- Supabase, Firebase, MongoDB

Deployment & IoT

- Git / GitHub, REST APIs
- Model + app deployment
- ESP32 / ESP8266, Arduino
- Leaflet.js, mapping tools

KEY ACHIEVEMENTS

- **Runner-up, CivicCode Hackathon 2026** — built Sunuwa, an AI-powered civic complaint routing platform, with the VagaBond team.
- **Runner-up, PCPS College Hackathon 2026** — recognized for an applied civic-tech build.
- **Led VagaBond** across multiple shipped civic-tech projects, owning both team direction and AI integration.
- **Built and deployed several end-to-end ML applications** spanning disaster risk, telehealth, and agriculture.

EXPERIENCE

Team Lead & AI Integration Lead — VagaBond2026 – Present

- Lead a three-person team building civic-tech and applied-AI projects for hackathons and real-world use.
- Own AI integration across projects — LLM pipelines, embeddings, clustering, and model deployment.
- Shipped multiple complete products, two of which placed as runner-up in college and national hackathons.

Independent Developer & ML Practitioner — Self-directed2023 – Present

- Design, build, and deploy full-stack and machine-learning projects end-to-end, from data and modeling to UI and deployment.
- Completed a self-paced AI/ML program and continue structured self-study in math, deep learning, and PyTorch.
- Focus on Nepal-specific problems: disaster resilience, public services, healthcare access, and agriculture.

PROJECTS

 **Sunuwa (सुनुवाइ):** AI civic complaint routing platform — citizens report issues, which are auto-classified, clustered, and routed. Built with Next.js, FastAPI, Supabase, Llama 3.3 (Groq), Gemini Flash, sentence-transformers, UMAP/HDBSCAN, and Leaflet.js. Runner-up at CivicCode Hackathon 2026.

 **Bāchā:** Photographs government Citizen Charters (नागरिक वडापत्र), extracts legally binding service commitments via Gemini Vision, and generates live service-deadline countdowns and auto-drafted compensation claims under Nepal's Good Governance Act.

 **QuakeSafe Nepal:** XGBoost model predicting earthquake building-damage grades on a 260k-building dataset — 74.45% accuracy, 0.70 macro F1 — using SMOTE and manual feature engineering, deployed via FastAPI with a web front-end.

 **Harvo:** Farmer-to-vendor marketplace with a Nepali voice interface and AI-based produce spoilage scoring, aimed at reducing waste and cutting out middlemen.

SajhaDoctor: Bilingual (Nepali/English) telehealth platform designed to improve access to basic medical guidance for rural users.

 **NepalPrep:** Entrance-exam preparation platform with structured notes, MCQ practice, and worked examples.

 **Smart Home IoT:** Award-winning ESP32/Arduino-based home automation system integrating sensors and controls.

ADDITIONAL INFORMATION

- **Tools & Tech:** React, Next.js, MERN, FastAPI, Flask, Python (scikit-learn, XGBoost), Tailwind, Framer Motion, Supabase, Firebase, Git/GitHub, ESP32/ESP8266/Arduino.
- **Languages:** English, Nepali, Hindi.
- **Certifications:** AI/ML microdegree (self-paced); ongoing self-study in linear algebra, deep learning, and PyTorch.
- **Education:** Higher Secondary (+2) completed. Preparing for BSc Computer Science entrance (2026 intake) during a gap year.
- **Interests:** Civic-tech, open source, applied ML for Nepal, hackathons.